

Derek Hollenbeck-Paniagua, PhD

1055 W Yosemite Ave, Apt 7, Merced, CA 95348

(209)898-5633 • dhollenbeck@ucmerced.edu

www.linkedin.com/in/dhollenbeck2

EDUCATION

PhD in Mechanical Engineering

May 2023

University of California, Merced

BS in Mechanical Engineering (high honors)

May 2016

University of California, Merced

AA in Mathematics & Physical Science

May 2012

Lassen Community College

RESEARCH EXPERIENCE

Postdoctoral Research Scholar

Dec 2023 – Dec 2025

40 Hours/Week

Center for Methane Emission Research and Innovation (CMERI) at UC Merced

- Lead the development and implementation of digital twin technologies to simulate and optimize methane emissions in various sectors, including energy production, agriculture, and waste management.
- Oversee the daily operations of the Center for Methane Emission Research and Innovation, including managing research projects, supervising graduate students and research staff, and coordinating collaborative activities.

Research Consultant

Mar 2023 – June 2023

SeekOps Inc.

10-20 Hours/Week

- Researched and developed localization algorithms using Digital Twins for point source emissions in an Oil and Gas setting.

Graduate Student Researcher

Dec 2021 – Oct 2022

CITRIS

10-20 Hours/Week

- Lead a team of researchers to design, develop, and demonstrate a hybrid VTOL fixed-wing aircraft for the inaugural CITRIS Aviation Prize. The aircraft developments included 4G telemetry, multi-point GCS communication, control handoff mechanisms,

edge-AI segmentation for road detection, OCN camera for residual dry matter estimation and a custom parachute.

Graduate Student Researcher

April 2021

Aerometix & Shell Global Solutions

10-20 Hours/Week

- Conducted field surveys using sUAS with miniature methane sensor at the Methane Emission Technology Evaluation Center (METEC). Lead a two man team to detect, localize, and quantify unknown emissions from sections of the METEC site to help evaluate different UAV-based quantification technologies.

Graduate Student Researcher

June 2019 – July 2019

Aerometix & PTAC

10-20 Hours/Week

- Conducted field surveys using sUAS with miniature methane sensor in several of Canada's oil and gas producer sites as a participant in Alberta Methane Field Challenge. Retrieved and compiled raw data to level 1 data sets for further analysis. Calculated release rates and locations of all the sites visited.

Graduate Student Researcher

Mar 2019 – Aug 2019

Honeywell

5 Hours/Week

- (NDA) Integrating new sensor onboard VTOL for vertical profiling of methane gas.

Graduate Student Researcher

Jun 2017 – Dec 2018

PRCI

10 Hours/Week

- Contributed to PRCI, NASA JPL, and UC Merced's efforts to integrate highly sensitive methane detection equipment on board fixed wing aircraft.
- Improving signal to noise issues by investigating vibrational and electrical noise.

Graduate Student Researcher

Aug 2016 – Dec 2017

NYSearch

10 Hours/Week

- Contributed to NYSearch, NASA JPL, and UC Merced's efforts to understand search and localization of methane leaks using highly sensitive methane detection equipment on board small unmanned aerial systems.

Graduate Student Researcher

June 2016 – Dec 2016

CITRIS

10 Hours/Week

- Studied airworthiness under supervision of the director of UC Center for Excellence on Unmanned Aircraft Systems Safety testing aircraft's flight performance and control system fail-safes.

Undergraduate Researcher

Spring 2016 – Fall 2016

Dept. of Mechanical Engineering

5-10 Hours/Week

- Worked closely with a graduate student and mentoring faculty to research and understand fundamental effects on nonlinear buckling dynamics of columns.

Undergraduate Robotics Prototyping

May 2015 – May 2016

Mechatronics, Embedded Systems and Automation Laboratory

5-10 Hours/Week

- Collaborated with graduate students on eDNA research designing UAS onboard multiple chamber fiber filter eDNA sampling system.

Mechanical Engineer Intern

June 2015 – Aug 2015

Ayda Technologies

40 Hours/Week

- Developed designs for manufacture and assembly while collaborating with product architect constructing product concepts and prototyping designs for beta testing.

TEACHING

ENGR 057 - Circuit Theory	Fall 2022
ENGR 057 - (IOR) Circuit Theory	Sum 2022
ENGR 065 – Statics and Dynamics	Fall 2021
ME 142 – Mechatronics	Spring 2020
ME 143 – Unmanned Aircraft Systems	Fall 2018
ENGR 057 – Circuit Theory	Sum 2017
ENGR 190 – Capstone Design	Spring 2017
ENGR 057 – Circuit Theory	Fall 2016

MENTORING

Postdoctoral Mentor, CH4Drone / HgDrone

Fall 2024 – Spring 2026

- Mentored a team of undergraduate students in developing a small unmanned aerial system equipped with a low cost methane sensor for real-time detection of methane leaks and mercury as a part of the DOE-EM grant.

Postdoctoral Mentor, BioGas Engine Project

Fall 2024 – Spring 2026

- Mentored a team of undergraduate students in developing a biogas engine capable of running unpurified methane gas for electrical generation using Freedom Motor's RotaPower.

Graduate Mentor, AIAA Design Build Fly

Fall 2016 – Spring 2020

- Mentored undergraduates by helping navigate rules and design problems to develop an aircraft to compete in the competition.

Project Manager, AIAA Design Build Fly

Fall 2015 – Spring 2016

- Lead, mentored, and guided a team of engineers towards the goal of designing, fabricating, and testing an electric remote-control aircraft to compete internationally.

PUBLICATIONS/CONFERENCE PAPERS/BOOK CHAPTERS

- **Hollenbeck, D.,** & Chen, Y. Smart Sensing with Digital Twins: Methane Emission Source Determination with sUAS. 2026. CRC Press
- **D. Hollenbeck,** R. Krzysiak, S. Cao, S. Giri, A. Saeed, P. O'Day, M. Beutel, and Y. Chen, "An optimal mobile sensing sUAS platform using Digital Twins and the observability Gramian," in 2025 13th International Conference on Control, Mechatronics and Automation (ICCMA). IEEE, 2025.
- Giri, S., **Hollenbeck, D.,** & Chen, Y. (2025, July). Which information metric is the best, CRLB, FIM or EMGR for optimal mobile sensing of a diffusing source?. In 2025 American Control Conference (ACC) (pp. 5166-5171). IEEE.
- **Hollenbeck, D.,** Edgar C., Euskirchen E., and Manies K. "Towards Mobile Wind Measurements Using Joust Configured Ultrasonic Anemometer for Applications in Gas Flux Quantification." Drones 9, no. 2 (2025): 94.
- **Hollenbeck, D.,** & Chen, Y. (2024, September). Towards A Digital Twin Framework for Landfill Emission Source Determination using Hybrid VTOL Fixed-wing: A Simulation Study. In 2024 20th IEEE/ASME International Conference on Mechatronic and Embedded Systems and Applications (MESA) (pp. 1-7). IEEE.

- An, D., Krzysiak, R., **Hollenbeck, D.**, and Chen, YQ. Long Endurance Site-Specific Management of Biochar Applications Using Unmanned Aircraft Vehicle and Unmanned Ground Vehicle. 2023 International Federation of Automatic Control. IFAC-PapersOnLine. 2023.
- An, D., Krzysiak, R., **Hollenbeck, D.**, and Chen, YQ. Battery-health-aware UAV mission planning using a cognitive battery management system. 2023 International Conference on Unmanned Aircraft Systems (ICUAS), 523-528. IEEE. 2023.
- **Hollenbeck, D.**, Zheng, K., Zulevic, D., and Chen, YQ. Swarm Robotic Source Seeking with Fractional Fluxotaxis. 2023 International Conference on Fractional Differentiation and Its Applications (ICFDA), 1-5. IEEE. 2023
- **Hollenbeck, D.** Digital Twin Enabled Collective Sensing and Steering for Source Determination Problems. Thesis. UNIVERSITY OF CALIFORNIA, MERCED. 2023.
- An, D., **Hollenbeck, D.**, Cao, K. and Chen, YQ. Soil methane emission suppression control using unmanned aircraft vehicle swarm application of biochar mulch-A simulation study. Journal of Information and Intelligence, 1, 1, 68-85. Elsevier. 2023.
- **Hollenbeck, D.** et al. Orange-Cyan-Near-Infrared Imagery Data for Management Unit 7 in Merced Vernal Pools and Grassland Reserve on September 27, 2022. Knowledge Network for Biocomplexity. 2023. (uuid_51675dd9-6d9b-45d6-a473-53566520270)
- **Hollenbeck, D.**, An, D., Krzysiak, R., and Chen, YQ. Towards Cognitive Battery Monitoring on Hybrid VTOL Fixed-Wing sUAS with Maximized Safe Endurance. In Proc. of the 2023 11th International Conference on Control, Mechatronics and Automation (ICCMA), 80—85. IEEE. 2023
- **Hollenbeck, D.**, Zulevic, D., and Chen, YQ. Single and Multi-sUAS Based Emission Quantification Performance Assessment Using MOABS/DT: A Simulation Case Study. 2022 18th IEEE/ASME International Conference on Mechatronic and Embedded Systems and Applications (MESA), 1-5. IEEE. 2022
- Manies, K., **Hollenbeck, D.**, Golston, L., Euskirchen, E. S., Edgar, C., Chen, YQ., and Waldrop, M. P. Methods for measuring wind speed and direction for accurate estimations of methane fluxes. AGU Fall Meeting Abstracts, A12E—04. AGU. 2022.
- **Hollenbeck, D.** and Chen, YQ. A Digital Twin Framework For Environmental Sensing with sUAS. Journal of Intelligent & Robotic Systems, 105, 1, 1-15. Springer Netherlands. 2022.
- Viola, J., Rodriguez, C., **Hollenbeck, D.**, and Chen, YQ. 13 A radio frequency impedance matching control benchmark and optimal fractional-order stochastic extremum seeking method. Outliers in Control Engineering. 2022. (Book Chapter)
- An, D, Liu, Z, **Hollenbeck, D.**, and Chen, YQ. A Greenhouse Gas Proximity Sensing System Using Chemiristive Strip and Miniaturized Radar Array. In Proc. of the 10th International Conference on Control, Mechatronics and Automation (ICCMA). 2022. (accepted)

- **Hollenbeck, D.**, Zulevic, D. and Chen, YQ. "A Modified Near-Field Gaussian Plume Inversion Method Using Multi-sUAS for Emission Quantification" In Proc. Of the 2022 International Conference on Unmanned Aircraft Systems (ICUAS). IEEE, 2022.
- An, D., **Hollenbeck, D.**, Gao, S., & Chen, Y. A Field Study of Soil Biochar Treatment Response Using Small Unmanned Aerial Systems (sUAS). In Proc. of 2022 International Conference on Unmanned Aircraft Systems (ICUAS) (pp. 1608-1613). IEEE. 2022
- **Hollenbeck, D.**, Zulevic, D. and Chen, YQ. "Advanced Leak Detection and Quantification of Methane Emissions Using sUAS." Drones 5.4 (2021): 117.
- **Hollenbeck, D.** and Chen, YQ. A Digital Twin Framework for Environmental Sensing with sUAS. J Intell Robot Syst. 2021
- Viola, J., **Hollenbeck, D.**, Rodriguez, C., and Chen, YQ. Fractional-Order Stochastic Extremum Seeking Control with Dithering Noise for Plasma Impedance Matching. Proceedings of the 2021 Conference on Control Technology and Applications. 2021.
- **Hollenbeck, D.**, and Chen, YQ. Digital Twin Behavior Matching of Gas Plumes Using a Fixed Sensor Mesh and Dynamic Mode Decomposition. In Proc. of the 2021 Symposium on "AI and Emerging Technologies for MESA (Mechatronics Embedded Systems and Automation)" of the 2021 ASME/IEEE International Conference in Mechatronics Embedded Systems and Applications (MESA) at the 2021 ASME IDETC/CIE, online.
- Fatehiboroujeni, S., **Hollenbeck, D.**, Mishra, A., and Goyal, S. Post-buckling Stability of a Cantilever Beam with Cubic Non-linearity in Constitutive Laws. Proceedings of the 17th International Conference on Multibody Systems, Nonlinear Dynamics, and Control (MSNDC). 2021
- **Hollenbeck, D.**, and Chen, YQ. MOABS/DT: Methane Odor Abatement Simulator with Digital Twins. 2021 IEEE International Conference on Digital Twins and Parallel Intelligence (DTPI). 2021
- **Hollenbeck, Derek**, and YangQuan Chen. "Multi-UAV Method for Continuous Source Rate Estimation of Fugitive Gas Emissions from a Point Source." 2021 International Conference on Unmanned Aircraft Systems (ICUAS). IEEE, 2021.
- Schloesser, Daniel S., **Derek Hollenbeck**, and Christopher T. Kello. "Individual and collective foraging in autonomous search agents with human intervention." Scientific Reports 11.1 (2021): 1-13.
- **Hollenbeck, D.**, Manies, K., Chen, Y., Baldocchi, D., Euskirchen, E. S., & Christensen, L. "Evaluating a UAV-based mobile sensing system designed to quantify ecosystem-based methane." AGU Fall Meeting Abstracts. Vol. 2020. 2020.
- **Hollenbeck, Derek**, and YangQuan Chen. "Characterization of ground-to-air emissions with sUAS using a digital twin framework." 2020 International Conference on Unmanned Aircraft Systems (ICUAS). IEEE, 2020.
- Whiticar, M., **Hollenbeck, D.**, Billwiller, B., Salas, C., & Christensen, L. (2020). Application of the bc GHGMapper™ platform for the Alberta Methane Field Challenge

(AMFC). Geoscience BC Summary of Activities 2019: Energy and Water, Geoscience BC, Report 2020–02, 87-102.

- Niu, H., **Hollenbeck, D.**, Zhao, T., Wang, D., & Chen, Y. (2020). Evapotranspiration Estimation with Small UAVs in Precision Agriculture. *Sensors*, 20(22), 6427.
- **Hollenbeck, Derek**, and YangQuan Chen. "A More Optimal Stochastic Extremum Seeking Control Using Fractional Dithering For A Class of Smooth Convex Functions." *IFAC-PapersOnLine* 53.2 (2020): 3737-3742.
- **Hollenbeck D**, Oyama M, Garcia A, Chen YQ. Pitch and Roll Effects of On-board Wind Measurements using sUAS. *IEEE Conf. Proc. ICUAS 2019*.
- **Hollenbeck D**, Dahabra M, Christensen LE, Chen YQ. Data Quality Aware Flight Mission Design for Fugitive Methane Sniffing using Fixed Wing sUAS. *IEEE Conf. Proc. ICUAS 2019*.
- Chen Y, **Hollenbeck D**, Wang Y, and Chen YQ. On Optimal Tempered Levy Flight Foraging. *Frontiers in Physics*. 2018. DOI: 10.3389/fphy.2018.00111
- **Hollenbeck D**, Schloesser DS, Kello KT. On Collective Behavior of Foraging Vultures and Individual Impact. *NSF Research Traineeship Annual Meeting 2018*. DOI: 10.13140/RG.2.2.20877.59369 (Poster)
- **Hollenbeck D**, Chen YQ. Fugitive Methane Detection and Localization with Small Unmanned Aerial Systems: Challenges and Opportunities. *CITRIS/CPAR Norcal Control Symposium 2018*. DOI: 10.13140/RG.2.2.15566.72004 (Poster)
- **Hollenbeck D**, Nunez G, Chen YQ, Christensen LE. Wind Measurement and Estimation with small Unmanned Aerial Systems (sUAS) Using On-Board Mini Ultrasonic Anemometers. *IEEE Conf. Proc. ICUAS 2018*. DOI: 10.1109/ICUAS.2018.8453418
- **Hollenbeck D**, Fatehiboroujeni S, and Goyal S. Effect of Non-Linear Constitutive Law on Buckling Dynamics. DOI: 10.13140/RG.2.2.18922.16322 (Poster)

AWARDS, FELLOWSHIPS AND GRANTS

DOE-EM (Key Personnel)	Fall 2025
“Development and Testing of Small Unmanned Aerial Systems for Real-time Detection of Methane Leaks and Mercury Emissions”	
DOE FOA 3256 Topic 3B -- (Senior Personnel)	Fall 2024
E2E-Methane-IQ	
DOE-EM – (Proposal Preparation)	Spring 2024
“Development and Testing of Unmanned Aerial Systems for Real-Time Methane Leak Detection”	
Center for Methane Emission Research and Innovation (CMERI)	Fall 2023
CITRIS Aviation Prize (inaugural)	Fall 2021
UC Merced ME Bobcat Fellowship	Fall 2021
UC Merced ME Bobcat Fellowship	Sum 2021

NSF NRT – Extended Intelligent Adaptive Systems Fellowship	Fall 2020 – Sum 2021
NSF NRT – Extended Intelligent Adaptive Systems Fellowship	Fall 2019
NSF NRT – Extended Intelligent Adaptive Systems Fellowship	Spring 2019 – Sum 2019
NSF NRT – Intelligent Adaptive Systems Fellowship	Spring 2017 – Sum 2018
NSF NRT – Interdisciplinary Computational Graduate Education	Spring 2017
Outstanding Student Award	Spring 2016

TECHNICAL SKILLS AND SOFTWARE

- MATLAB, Simulink, Mathematica, NetLogo
- Computer aided design/engineering
- CNC Machining
- Mechatronics and embedded systems
- Control Systems, Fluid Mechanics, Dynamics
- Computational Fluid Dynamics
- FAA Part 107 Certified

ATTENDED CONFERENCES / WORKSHOPS

- 2025 ICCMA, Paris, France
- 2024 MESA, Genoa, Italy
- 2023 CH4 Connections, Fort Collins, Colorado.
- 2023 International Conference on Fractional Differentiation and It's Applications (ICFDA).
- 2022 18th IEEE/ASME International Conference on Mechatronic and Embedded Systems and Applications (MESA)
- International Conference on Unmanned Aircraft Systems, Dubrovnik, Croatia (ICUAS) 2022
- International Design Engineering Technical Conferences & Computers and Information in Engineering Conference, virtual (IDETC-CIE) 2021
- International Conference on Unmanned Aircraft Systems, Athens, Greece (ICUAS) 2021
- Digital Twins and Parallel Intelligence, Beijing, China, (DTPI) 2021
- Workshop: Nonlocal Models: Analysis, Optimization, and Implementation, Trier, Germany 2021
- International Conference on Unmanned Aircraft Systems, Athens, Greece (ICUAS) 2020
- International Federation of Automatic Control, Germany (IFAC) 2020
- American Geophysical Union, virtual (AGU) 2020
- NSF Spring Retreat 2019
- NSF Research Traineeship (NRT) Annual Meeting 2018

- Fractional PDEs: Theory, Algorithms and Applications 2018
- International Conference on Unmanned Aircraft Systems (ICUAS) 2018
- CITRIS/CPAR Control Theory and Automation Symposium 2018
- Modeling, Analysis and Numerics for Nonlocal Applications (MANNA) 2017
- International Congress of Theoretical and Applied Mechanics (ICTAM) 2016

PRESENTATIONS

ME 190 (Digital Twins)	Fall 2024
(title: Smart Environmental Sensing with Digital Twins)	
PhD Dissertation Defense	Spring 2023
(title: Digital Twin Enabled Collective Sensing and Steering for Source Determination Problems)	
ICFDA 2023	Spring 2023
(title: Swarm Robotic Source Seeking with Fractional Fluxotaxis)	
MESA 2022	Sum 2022
(title: Single and Multi-sUAS Based Emission Quantification Performance Assessment Using MOABS/DT: A Simulation Case Study.)	
ICUAS 2022	Sum 2022
(title: A Modified Near-Field Gaussian Plume Inversion Method Using Multi-sUAS for Emission Quantification)	
CITRIS Research Exchange	Spring 2022
(title: Long Endurance Edge-AI Platform For Research Opportunities and Data Gathering)	
PhD Dissertation Qualifying Exam	Spring 2022
(title: Digital Twin Enabled Collective Sensing and Steering for Source Determination Problems)	
MESA Lab Research Group Meeting	Fall 2021
(title: The empirical gramian and why it is important)	
MSNDC 2021	Sum 2021
(title: Post-buckling Stability of a Cantilever Beam with Cubic Non-linearity in Constitutive Laws)	
ICUAS 2021	Sum 2021
(title: Multi-UAV Method for Continuous Source Rate Estimation of Fugitive Gas Emissions from a Point Source)	
DTPI 2021	Sum 2021
(title: MOABS/DT: Methane Odor Abatement Simulator with Digital Twins)	
MESA Lab Research Group Meeting	Spring 2021
(title: Swarm Robotic Source Seeking with Fractional Fluxotaxis and Seasonal Meandering)	
MESA Lab Research Group Meeting	Fall 2020
(title: Physicomemetics, Chemical Plume Tracing, and Fluxotaxis (I))	
MESA Lab Research Group Meeting	Fall 2020
(title: MOAB/DT: Towards Smart Swarming for Methane Emission Sensing and	

Understanding/Quantification)	
ICUAS 2020 Workshop Tutorial	Sum 2020
(title: A digital twin framework on using drones as edge devices for environmental monitoring)	
ICUAS 2020 Workshop Tutorial	Sum 2020
(title: Methane sensing drones with machine learning)	
UC Agriculture and Natural Resources (UCANR) DroneCamp	Spring 2020
(title: Methane emission drone swarm sensing to better understand permafrost thawing dynamics.)	
MESA Lab Research Group Meeting	Fall 2019
(title: Fractional Order Modeling of Atmospheric Dispersion)	
MESA Lab Research Group Meeting	Spring 2019
(title: UAV Gas Sensing, a Digital Twin, and PET (POSIM Experimental Testbed))	
Invited Talk – Applied Math Optimization Research Group	Spring 2019
(title: tbd)	
NSF NRT Retreat Meeting	Spring 2019
(title: Collective Foraging Among Loosely Coupled Agents)	
Invited Talk – NSF NRT Fall Kickoff Meeting	Fall 2018
(title: On Collective Behavior of Foraging Vultures and Individual Impact)	
NSF NRT Retreat Meeting	Spring 2018
(title: Collective Foraging Among Loosely Coupled Agents)	
Invited Talk – Applied Math Optimization Research Group	Fall 2017
(title: Fractional Calculus Enhanced Source Seeking of Fugitive Methane)	
Guest Lecture ME190 –	Sum 2017
(topics: Communication Basics, Long Distance Drones, UTM)	
Guest Lecture ME190 – Long Distance Drones	Spring 2017
Guest Lecture ME190 – Long Distance Drones	Fall 2016
Guest Lecture ME190 – Unmanned Aerial System Traffic Management	Fall 2016